

A* SEARCH ALGORITHM

Real-time pathfinding

A*

Heuristic comparison

Obstacle grid navigation

Step-by-step exploration

COST FUNCTION

$$f(n) = g(n) + h(n)$$

Total cost = Cost from start + Heuristic to goal

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WHAT IS A* ?

A* (pronounced "A-star") is a best-first graph search algorithm that finds the shortest path between two nodes.

Introduced by Hart, Nilsson & Raphael (1968), it is complete and optimal — guaranteed to find the shortest path if one exists.

Unlike Dijkstra (which explores uniformly outward), A* uses a heuristic $h(n)$ to guide search toward the goal, dramatically reducing nodes explored.

ALGORITHM STEPS

1. Add start node to Open List with $g = 0$
2. Pick node with lowest $f(n)$ from Open List
3. If it's the goal → trace path and stop
4. Move current node to Closed List
5. For each neighbor: skip if wall or closed
6. Compute $g' = g(\text{current}) + 1$
7. If $g' < g(\text{neighbor})$, update g , h , f , parent
8. Add/update neighbor in Open List
9. Repeat from step 2

COMPLEXITY

Time $O(b^d)$ b = branch factor, d = depth**Space** $O(b^d)$

All nodes stored in memory

Best $O(d)$ Perfect heuristic $h = h^*$ **Worst** $O(b^d)$ $h(n) = 0$, Dijkstra mode

Complete

Optimal

Admissible $h(n)$ required

OPEN & CLOSED LISTS

- Open List holds discovered nodes not yet fully processed. Sorted ascending by $f(n)$.
- The node with lowest $f(n)$ is always chosen next — the "best-first" strategy.
- Closed List holds fully explored nodes whose optimal cost $g(n)$ is confirmed.
- Nodes in Closed List are never re-expanded, guaranteeing efficiency and no cycles.
- If Open List empties without reaching the goal, no valid path exists in the graph.

HEURISTIC FUNCTIONS

- $h(n)$ must be admissible — never overestimate the true cost. This guarantees optimality.
- Manhattan: $h = |dx| + |dy|$ — best for 4-directional grid movement.
- Euclidean: $h = \sqrt{dx^2 + dy^2}$ — best for any-angle movement.
- Chebyshev: $h = \max(|dx|, |dy|)$ — best for 8-directional grid movement.
- $h(n) = 0$ reduces A* to Dijkstra's algorithm (exhaustive, always optimal but slower).

APPLICATIONS

GPS Navigation

Google Maps, Waze, TomTom use A* for fastest driving routes.

Game AI

NPC movement in Warcraft, StarCraft, The Sims — real-time game AI.

Robotics

Autonomous drones and warehouse robots for obstacle-free navigation.

Bioinformatics

Protein folding path search and genome sequence alignment.

PSEUDOCODE — A* Algorithm

```
function A_Star(start, goal):
    openList = {start}
    closedList = {}
    start.g = 0
    start.h = heuristic(start, goal)
    start.f = start.g + start.h
```

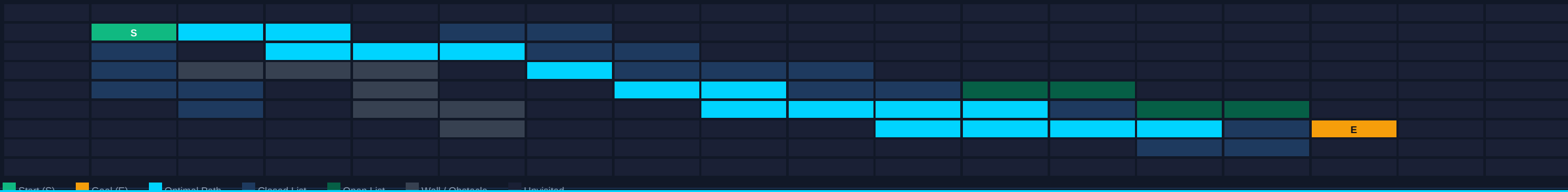
```
while openList is not empty:
    current = node with lowest f(n)
    if current == goal:
        return reconstruct_path(current)
    move current to closedList
```

```
for each neighbor of current:
    if neighbor in closedList or
       is_wall: continue
    tentative_g = current.g + 1
    if tentative_g < neighbor.g:
        neighbor.parent = current
        neighbor.g = tentative_g
        neighbor.h = heuristic(...)
```

ALGORITHM COMPARISON

Algorithm	Optimal	Complete	Speed
A* Search	✓ Yes	✓ Yes	★★★★★
Dijkstra's	✓ Yes	✓ Yes	★★★★
Greedy BFS	✗ No	✗ No	★★★★
BFS	✓ Yes	✓ Yes	★★★
DFS	✗ No	✗ No	★★★★

GRID SIMULATION — Sample A* Pathfinding Run



Start (S) Goal (E) Optimal Path Closed List Open List Wall / Obstacle Visited

Artificial Intelligence

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